ME101: Engineering Mechanics, 2023-24

IIT Guwahati

Tutorial – 11

Problem1:

A 10-kg wooden plate is suspended from a pin support at A and is initially at rest. One 5kg metal sphere is released from rest at B and falls into a hemispherical cup C attached to the plate at the same level as the mass center G. Assuming that the impact is perfectly plastic, determine (a) the angular velocity of the plate and (b) the velocity of the mass center G of the plate immediately after the impact. You can consider dimensions L_1, L_2 and L_3 as 0.8m, 1.6m, and 0.5m respectively. [10 Marks]

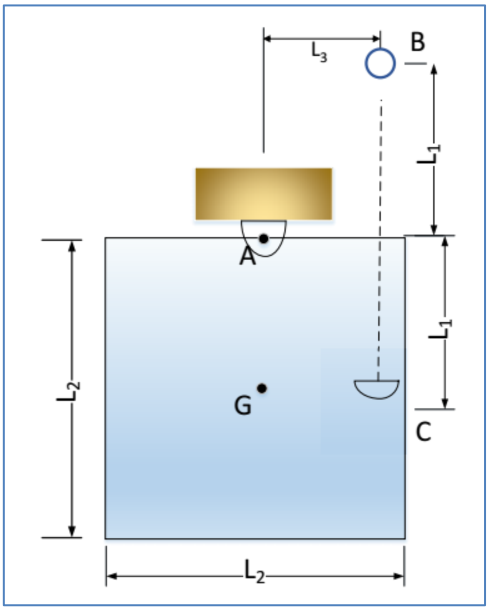

Figure 1: Problem1

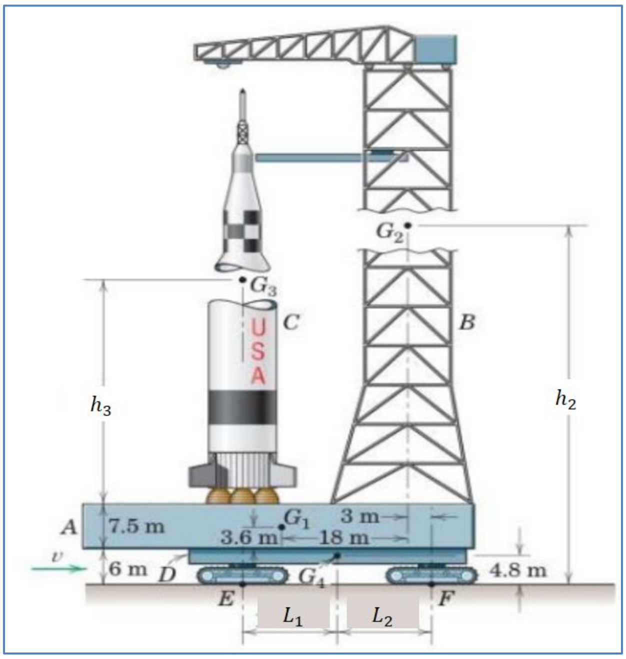

Figure 2: Problem2

Problem 2:

Figure 2 shows the Saturn V mobile launch platform *A* together with the umbilical tower *B*, unfueled rocket *C*, and crawler-transporter *D* which carries the system to the launch site. The approximate dimensions of the structure and locations of the mass centers *G* are given. The approximate masses are $m_A = 5$ Gg, $m_B = 7$ Gg, $m_C = 0.5$ Gg, and $m_D = 3$ Gg. The minimum stopping distance from the top speed of 3 km/h is 0.15 m. Compute the vertical component of the reaction under the front crawler unit F during the period of maximum deceleration. Considering $h_2 = 80$ m, $h_3 = 50$ m, $L_1 = L_2 = 15$ m. [10 Marks]

Problem 3:

The block *A* and attached rod have a combined mass of 10 kg and are confined to move along the 60° guide under the action of the 120 N applied force. The uniform horizontal rod has a mass of 4 Kg and length 0.6 m is welded to the block at *B*. Friction in the guide is negligible. Compute the bending moment *M* exerted by the weld on the rod at *B*. [10 Marks]

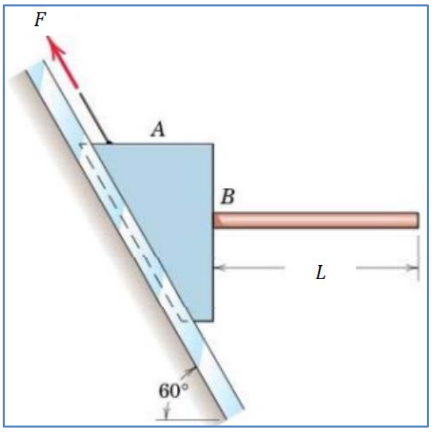

Figure 3: Problem 3

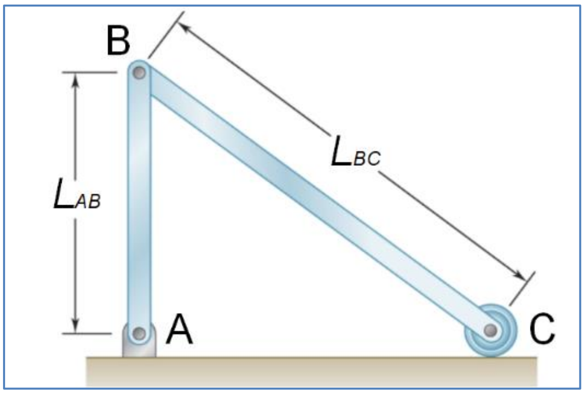

Figure 4: Problem 4

Problem 4:

The uniform rods AB and BC of masses 10 kg and 15 kg, respectively, and the small wheel at *C* is of negligible mass. Knowing that in the position shown the velocity of wheel C is 10 m/s to the right, determine the velocity of pin B after rod AB has rotated through 90°. Here $L_{AB} =$ 1.2 m and $L_{BC} = 2$ m. [10 Marks]

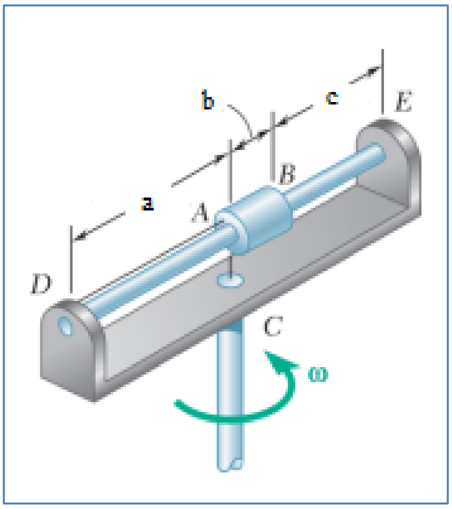

Figure 5: Problem 5

Problem 5:

A 1 kg tube *AB* can slide freely on rod *DE*, which in turn can rotate freely in a horizontal plane. Initially the assembly is rotating with an angular velocity ω = 10 rad/s and the tube is held in position by a cord. The moment of inertia of the rod and bracket about the vertical axis of rotation is 0.50 kg.m^2 and the centroidal moment of inertia of the tube about a vertical axis is 0.003 kg.m^2. If the cord suddenly breaks, determine (a) the angular velocity of the assembly after the tube has moved to end *E*, (b) the energy lost during the plastic impact at *E*, (c) sliding velocity of tube *AB* just before the impact. Assume dimensions are a = 300 mm, b = 100 mm, c = 200 mm. [10 Marks]